Y Combinator CEO on How AI is Changing What Startups Build

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AI Executive Summary

1. Core Thesis: Gary Tan argues that AI is fundamentally accelerating startup creation and productivity by shifting the burden from execution to strategic intent; while technical tasks can be automated in 6–18 months, human oversight remains critical because it involves understanding context and defining what problems are worth solving. He advocates for democratizing access to advanced models (including open-source or international ones) rather than centralizing power with Big Tech firms, suggesting that societal benefit will grow as AI allows small teams to execute work previously requiring large organizations.

2. Key Arguments & Data:

3. **Startup Viability:** Y Combinator receives ~100,000 annual applications and invests up to $500,000 per founder; recent startups have gone from zero to tens of millions in revenue with fewer than 20 people (vs. historical norms requiring hundreds/thousands for similar funding).
4. **Productivity Scaling:** Compared to Tan's early YC years where he wrote ~15–20 lines of code/day, his productivity today using AI agents is estimated at roughly 40 times more output; effectively one person now performs the work previously done by 10 to 100 employees.
5. **Timeline for Automation:** Tan predicts coding agents will perform tasks equaling his job in six to 18 months (e.g., he stopped drinking alcohol to aid focus and found it difficult). He would return to moderation only when AI no longer needs him as an intermediary ("coding agent don't need me anymore").
6. **The Human Element:** While mechanics get easier, humans are needed for "knowing what to build," understanding context, interpreting desire from other humans, and knowing *why*. Tan views the role of leadership in this new age not as execution but as orchestration (the conductor analogy).
7. **Hard Tech & Investment Focus:** YC is shifting funding focus toward hard tech (nuclear reactors, defense) rather than traditional software; cites Torinol mosquitoes drone killer using computer vision/sensors to

demonstrate AI enabling tangible physical solutions previously deemed illegal or cost-prohibitive.

8. **Democratization vs. Monopoly:** Argues against Big Tech controlling APIs/models and platforms for proprietary reasons without competition; supports open-weight models (including Chinese) if safety guardrails are applied ("users should have choice" over centralized control).

9. **Organizational Inertia:** Corporate AI transitions in Fortune 500 companies are slower than expected due to organizational latency, which Gary views as a "societal saving grace." While individual productivity increases (some work twice more), the transition allows society time to adapt.

10. Actionable Takeaways:

11. **Leverage Agents with Care:** Adopt personal AI agents for research and writing tasks but never grant them write access or permission to press send on sensitive communications without human review ("watchers of what your open claw does").
12. **Focus on Strategy, Not Just Code:** As technical barriers lower, founders should focus their ambition on identifying "what people want" and solving problems rather than optimizing execution speed. Aim for the 40x productivity multiplier but retain human oversight.
13. **Support Open Models & Competition:** Encourage or invest in open-weight AI ecosystems to prevent Big Tech entrenchment; competition drives better consumer choice regardless of model geographic origin if safety is guaranteed via "guardrails."
14. **Embrace Hard Problems:** Look beyond software for opportunity (nuclear energy, agriculture/defense); use AI agents to make tangible, expensive physical problems viable with smaller teams.

Full Transcript

If you want to know how AI is going to change the workplace, one of the best places to look is at startups. And there's probably no one more steeped in the world of startups than this guy. Hi, my name is Gary Tan, and I'm the president and CEO at Y Combinator. If you haven't heard of Y Combinator, you've definitely heard of some of the startups it's helped launch. Airbnb, DoorDash, Coinbase.

YC, as it's commonly known, is a startup incubator. Gary says it gets about 100,000 applications every year, but only a few startups get accepted. Those that are get up to a half a million dollar investment. Gary prolifically posts about his AI use, and he says that AI is changing everything about how companies are being created these days, and it's also changing how he works as a CEO. Last month, I got on a call with Gary. He was in San Francisco and I was in a studio overlooking Times Square. If you want to see a video of our conversation, head to Spotify. You use AI agents a lot. You've talked a lot about how ways you're putting it into your life. What are you having it do and what do you find that it's most useful at? I mean, I think of it as like self-driving rocket for the mind. And so... Interesting. Like, back in the day you might have to have a team of a hundred or a thousand people to get done something that now only takes like...

10 or 15 people. And so I think that incrementally, this means that we need to boil the ocean. We need to increase the ambition of founders. And there are just more things that people should be doing. How long do you think it'll be until AI is as good at your job as you are? It can't prompt yet. So I'm on the lookout for that. I actually stopped drinking in February because I found that, you know, even having a single beer would make it much, much harder to think about code and like sort of run coding agents. But, you know, I would say I'm going to start drinking again when the coding agents don't need me anymore. And I would guess that sometime in the next like...

Six to 18 months. Wow, that's soon. Yeah, I mean are you worried though about your own? I think the most important thing at that point will still be the most important thing it'll be knowing what to build It'll be like you still

need a human. You still need a human who knows other humans You still need to understand context. You still need to figure out Well, what do people want and why and then how do I build a thing? The mechanics of it get easier and easier, but knowing what to build and knowing what to want is still a very, very human thing. Welcome to The Journal, our show about money, business, and power. I'm Ryan Knudson. Coming up on the show, a conversation with Gary Tan about AI and startups. I've just recorded the first episode of our new season talking about the state of play for the IMO's Net Zero framework.

in the wake of the just concluded latest MEPC meeting. The big question I hear again and again is, what happens next and how quickly? In this episode, I unpack what's moving, what's stalling, and what that means for future decisions and timelines. Search for Maritime Impact on your favorite podcast platform. Stay tuned. two extreme views of where AI might take us. On the one hand, there's people that worry about an AI apocalypse that it might bring about the end of humanity. And there is, on the other end of the spectrum, there are people who think that it could bring about a utopia in the future. Where do you fall on that spectrum? I mean, I think it's going to be different. And utopia is probably too, like, things are going to change. And anytime the cheese is moved, there are going to be winners and losers.

And then net net, my hope is that there are more winners than losers because it's just a really, really powerful technology. And from our perspective, you know, we're trying to help people start new businesses. And I deeply believe that doing that right means that you're solving problems for other people. And so in aggregate, like. you know, this kind of technology in particular makes that way, way more possible. And so will people lose jobs? Yes. On the flip side though, like with AI, it's like, you know, you used to have to hire a lot of people and raise a crazy amount of money. And, you know, now you're actually able to create businesses for far, far less. And there can be a lot more of them. So in aggregate, like I think Humans can solve more problems and there's a lot more entrepreneurship that can happen like our standard of living will rise and Things will just actually be the way they should be and maybe that is utopian So obviously why combinator you invest in startups you nurture startups in what ways are you seeing? AI allow people to start startups more

quickly I mean the most simple version of this is you're able to write like one person is able to write a large multiple of what they used to be able to do. I run YC today. About 13 years ago, I was a designer in residence and I was building software for YC. In those days, I was writing about 15 to 20 lines of code a day. That's not a lot, but now with agents, I've been doing something closer to 400 times that.

you're able to, one person is actually literally able to do the work of 10 to 100 people previously. And so what that means, we're sort of seeing dozens of companies in the last few years go from zero to tens of millions of dollars in revenue with like less than 20 people having only raised a seed round of like under five or 10 million. So a lot of this is like unprecedented. Like you've never seen people be able to do that before. And this is great. Like this is all great news for people who want to start companies. It's great for people who want to start companies. Maybe not so great for people who want to work at those companies, but in what, how much- But those people should start companies. How much do you think that that can scale? As a business gets bigger and bigger and bigger, at what point does it need to start hiring more people or will it ever? I mean, this is where I think this requires a change in mindset, right?

If you just want to maintain things as they are, you just want to sort of let people go. But I think that the world is actually so full of problems that we're entering this other age. And this might be a change in the nature of types of companies that can be started. Gary says that the types of companies that Y Combinator had traditionally funded were software companies. But thanks to AI, YC is now working more in hard tech. things like nuclear reactors and defense. We have a company called Torinol that kills mosquitoes using drones, for instance. Did I see a video of them recently where there's a little tiny drone that can... What does it do? Is it like... I don't know how... Yeah, it's got computer vision, it has a bunch of sensors. And it just zaps mosquitoes. Oh, it just flies right through them. And just kind of cuts them up. Yeah. And so you have an array of them in your backyard and...

You just don't have mosquitoes anymore. That would be nice. There's a very concrete version of this. Five or 10 years ago, this is a company that maybe

wouldn't have raised money. It just wouldn't have been too expensive. What if someone wants to make a company and they think of us as the mosquitoes? I mean, I'm being glib. Well, then it'd be illegal. Right. Then it'd be illegal. I'd be the first to say, if this is starting to... destroy society or you know be deleterious like hey maybe this is something we should slow down but I just see too many I see too many places where this is actually so useful and so powerful and so empowering you know here's a startup that if the cost of doing it became viable if it if you can do it like better faster cheaper and then it becomes something that is ubiquitous like you know Would love to be able to go outside and not be covered in bug bites are there certain kinds of companies people that have startup ideas that you're gonna that you say we're not gonna fund that or work with that because we think that that idea of yours While interesting is maybe too vulnerable to AI Honestly everything is so brand new like the rulebook has been thrown out like three or four years ago You could argue that there were sort of things that thousands of founders have gone after and tried to do. And you could just safely say, no, that's not going to work. Like a thousand people have tried that. But right now, I think we're getting into a moment now where it's like, actually, all the rules have been rewritten and everything is new again. What about your company, Y Combinator? How much are you relying on AI to make decisions that a human used to make? Every single person at YC now has their own personal AI agent.

We have a company, Brain, that actually flattens access to all the information so that it's actually super democratized. Have you run into any problems with that? I mean, sometimes agents can run amok a little bit and do things that you didn't intend for them to do. Oh yeah. I mean, the thing is that you can just have watchers on that. There's actually a huge cottage industry now of like, how do you make software that is watched by other AI? And so we funded a company called Clawvisor, for instance, that is sort of a watcher of what your open claw does. And so if you want to access Gmail or make a draft, it itself is an AI that is watching that AI. And saying, hey, and if it starts to stray from its objectives in any way. Then it sends you a message. It's like, hey, looks like it was trying to do this. Is this what you intended or not? Interesting.

I mean, you always hear about these stories about, oh, the agent went and deleted my whole hard drive and things like that. I mean, we tend to be very careful about write access. And then when I'm actually sending a text, when I'm sending email, when I'm sending Slack. People often joke these days. It's like, oh, Gary, is that you? It's like, yeah, yeah, it's all me. If I send you an email, it's me. If I send a... You never let your agent actually press send on any message. That's right. Yeah, I have to... I copy and paste and I send it myself. I feel like that's my covenant with people out there. After the break, more with Gary Tan on the less utopian side of AI. I've just recorded the first episode of our new season.

talking about the state of play for the IMO's Net Zero framework in the wake of the just concluded latest MEPC meeting. The big question I hear again and again is, what happens next and how quickly? In this episode, I unpack what's moving, what's stalling and what that means for future decisions and timelines. Serve for maritime impact on your favorite podcast platform. Stay tuned. So there's a growing backlash to AI in some corners. You're seeing a lot of opposition to data centers. You're seeing even executives of AI companies face threats of physical violence. How do you talk about that issue with your startup founders? And what do you make of that? I mean, we'd certainly have a responsibility to the people when we build these companies, right?

Are we taking care of people's data? Are we trying to be benevolent? Are we trying to create something that actually solves legitimate problems? And if you do that, that's great. I mean, I think AI is the most powerful new technology, obviously. And it can do a lot of great and a lot of good in the world. A lot of people worry that it won't. Right. No. I mean, how do you bring more people on board and, you know, how do you try to solve this issue where you're getting people who, you know, are dedicating their lives in some cases to try to stop the development of this technology? I mean, I wish they would just use the technology themselves, right? Like, there's new innovation that is happening, whether it's math, physics, I mean, they're just... I mean, that's powerful. And that's pushing forward what humans are actually...

capable of doing. And what I hope is that that translates into like real medical breakthroughs, for instance, like being able to solve cancer, being able to actually make way on clean energy. Do you think it's going to take a breakthrough like that before more people get excited or to... I mean, it's happening, it will happen. And I think the fear is not totally unfounded. I mean, even the most accelerationist people I know are worried about, you know, Ken Skynet happened, but you know I love movies from the 90s and 2000s as well But that's not the only world right I mean that's the premise of the Terminator obviously is go back in you know Terminator 2 go back in time try to stop this company from making this technology Does that keep you up at night at all? I mean that's a movie and it's it's sci-fi well sure that's a movie but I mean there are people you know there you know someone through Molotov cocktail at Sam Altman's house. I mean like right does that does that concern you on a personal level that absolutely I mean, I think that you know both political and I mean, it's basically another form of political violence We already have laws that constrain the things that are deleterious in society I'm not saying that we don't need new laws eventually, but it's so premature It's so early and right now like we're in the steep part of the curve to scaremonger and scare people away from the super powerful technology is just, it's sad. Instead of trying to stop it, use it. And so if we took even half of the energy that's going into sort of scaremongering about AI and actually just using it, let's solve these problems. There are just too many out there to solve. And that's the world I want to live in. Gary is very active in local politics in San Francisco and California.

One of the things he cares most about when it comes to regulating AI is making sure everyone has access to the best models, which right now are controlled by open AI, anthropic, and Google. One of the things that we've been tracking is sort of this fight between big tech and little tech. And this always happens. You know, you can look at an Apple or even a Google and say, well, as time goes on, They want to capture more and more revenue, so they will actually close their platforms to others. And that's probably the worst thing that can happen to tech and to tech innovation. And so that's probably the number one thing that I'm most focused on. I mean, obviously, Y Combinator is unique in that we're sort of like the defining union or guild for early stage startups.

And the thing that allows them to succeed or kills them ultimately is access to markets, platforms, APIs, and ultimately LLMs. And so we really, really care about access to all of those things. Access meaning that you don't want Open Air, or Anthropocrania, or any other company to put up some rule that says you can't use our technology for X, Y, and Z reason or that it should be a certain price. Yeah. Users should have choice. And that choice actually happens through lots and lots of small startups, all offering better and better things that are in competition with one another. That's how consumers get choice, and that's how technology should work. And it certainly isn't how it works yet. So are you saying that you think that the big AI companies now, should they have open source versions of their models? Because now, the way things are going, it seems like this will entrench.

the big tech companies at the center of the economy and give them an enormous amount of power. If there's suddenly all new businesses are being built off of their LLMs, how do you avoid that future? I mean, ultimately, I think there's a debate happening right now in the halls of power in DC about whether or not people should be allowed to have access to Chinese open-weight models. And I would argue very much in favor of... That people should have access to that like that is probably the biggest democratizing factor for Why because it gives more competition to American companies or what's the what's the argument? It just gives competition period it gives choice and so I guess there's two things One is that the LLM itself is turning into a little bit of a commodity like we'll sort of switch to whatever the best one is and we sort of hope that that remains an American company and then there's a lot of tension right now around you know as we talk right now like Kimmy just came up with a Chinese model just came out with and they're about to release open weights on something that is almost as good as the very best American models. Would you switch to a Chinese model?

With the right guardrails, I would because you can actually look at, you can instrument everything about what goes in and out of those models. And, you know, once again, like all AI can be safe enough if you watch it and if you put it in the right harness and watch it the right way. Why does it matter than if those are American companies? I think that intelligence is so important that like it's really about access, right? If we want American

companies, we want American innovators, and we want American defense companies, for instance, to be able to use whatever the absolute best is. When you're ultimately talking about dominance at the societal level, whose companies are the most innovative, whose products are the best, whose military is the best.

over the next course of decades, all of those will ultimately be answered by who has the most intelligent models. For the past few years, the headlines on AI have been all about the inevitability of mass layoffs. It's something we've covered a lot here on The Journal, and we hear about it from listeners a ton. But it turns out that the transition to AI at the corporate level is happening more slowly than expected. And Gary thinks that's a good thing. Society will absorb this. I think there will be transformation in every organization, but it'll happen slower than we might think, and that might be a good thing. Like when you look at a Fortune 500 company and you think, oh, it's going to take forever, like that's not a bug. It might actually be a societal feature, right? The organizational latency, just sort of the inertia of big companies it may well be the saving grace for society at large. I interviewed somebody recently who compared AI to being a conductor and he was like, you know, he worked in marketing and he was saying how in marketing, you know, you used to just think about, okay, I want to play the drums and that's, you know, email communications or I want to work in strategy and that's you know, playing the flute and now everybody needs to think of themselves more as conductors because they can use AI to do all that. And I thought that was a really interesting analogy, but I had also made me think that, you know, not everybody wants to be a conductor. Some people do just want to like wake up, put on their, you know, put on their shoes and go hit the triangle. And so like, what is that going to do to society if people have to, if in order to succeed, you have to level up in terms of how you even think about what a job is. And a lot of people may not want to do that.

I mean, it seems like we need to be more creative about what these jobs entail and look like. And on the flip side, maybe that's where live events come into play. It's possible that we should be throwing a thousand times more parties. I think that what are people going to do with all their time? They can actually focus on the arts. I mean, I always think about, Aldous

Huxley talked about in the 20th century that... You know, why are we doing 40 hour work weeks? Like maybe it should be 20. Like these things become more and more possible as the standard of living can rise. All technological innovations up until this point have not really resulted in less time working though. Why do you think that this one might? Right. It's... I mean, what's funny is I sort of know for a fact that most people who become very AI-pilled, myself included, like work about...

two times more. Yeah, I feel like I've heard that from a lot of people. Once they realize what they can do, they like can't stop. They're like, oh my God, I can do this and I can do that. And they just like it starts to infill up their life as they start to do more and more with it. But that's obviously you have to have that sort of drive inside yourself, you know. That's right. I mean, it might just end up, you might just end up having people who are like very, very productive and other people who don't have to be right. Like my hope is that It unleashes a brand new just a firestorm of entrepreneurship of like there are so many problems out there that we could solve for one another And then maybe they don't have to be all billion dollar companies like they might be Like actually a really nice bar where you have community like maybe it's a return to third spaces I mean there are just so many things that people like and enjoy that going back to YC, is YC going to be run by one person and a whole bunch of AI agents? The reality is like, no, never. We want to bring together really smart founders who have been there with the next generation of people to do that. And so the people are the point. And if I expand that out into broader society, it's like we should have better restaurants. We should have better social services. We should have better cities.

we should have more housing. Like, what are the things that people actually want? Let's work backwards from that. And more and more people can go towards that. We will never truly achieve utopia, but that doesn't mean that we can't try. Well, thanks so much for your time, Gary. I really appreciate it. Thanks so much for having me. Great questions. That's all for today. The Journal is a co-production of Spotify and the Wall Street Journal. If you like our show, follow us on Spotify or wherever you get your podcasts. throughout every weekday afternoon. Thanks for listening. See you

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